PHYS7101: NUMERICAL METHODS AND DATA ANALYSIS · LECTURE 7

From series to shapes

Phase space, correlation and PCA, and other decompositions

Mathew Smith & Aneta Stefanovska

School of Physics and Astronomy | Lancaster University

Where we left off

From earlier lectures

- Limit cycle: a self-sustained oscillation (L2)
- Eigenvalues and eigenvectors (second-year linear algebra)
- The Jacobian matrix of a system (L3)
- Autocorrelation; the order carries the information (L6)

Today

A Phase space: the third view of a time series; fixed points, limit cycles, eigenvalues

B Decomposing data: correlation, SVD, PCA

C Outlook: other decompositions

Lecture 6: two views of a time series, in time and in frequency. Today the third: phase space.

Each decomposition assumes something about the shape of the set of points: look at the points first. Running example: the van der Pol oscillator of Lectures 2 and 6, $\mu = 2$.

PART A

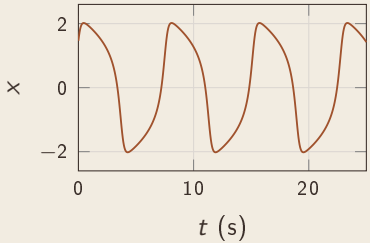
Phase space

A series as a path through the states of the system.

One series, three views

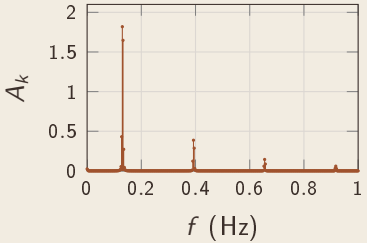
Problem: a plot of $x(t)$ shows one value at a time. How do we see the state of the system?

time domain (L6)



period, shape, slow and fast parts

frequency domain (L6)



$f_0 = 1/P$ and its harmonics

phase space (today)



the state: one closed curve

The same van der Pol series, $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$ with $\mu = 2$, in all three. Right: each moment plotted as velocity $\dot{x}$ against position x – every period traces the same closed curve.

Data: van der Pol, $\mu = 2$, RK4 (L1, L2), every 0.05 s; used throughout, noise added where stated.

State, phase space and trajectory

Problem: what do we need to know at one moment to predict the future?

- **State:** the numbers that, with the equations, fix the future (for a deterministic system; with noise, only its probabilities). For a second-order equation such as van der Pol: position and velocity, $(x, \dot{x})$.
- **Phase space:** the space of all possible states; here the plane $(x, \dot{x})$.
- **Trajectory:** the path of the state in time, starting from the initial condition (Lecture 1: an initial-value problem).

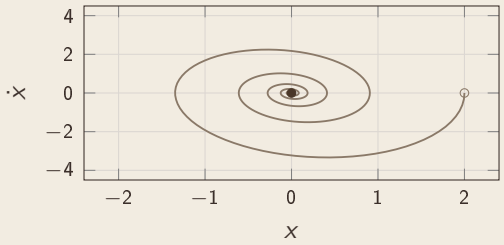
Autonomous and nonautonomous

Autonomous system (L2): through each point of phase space passes exactly one trajectory – trajectories never cross.
Nonautonomous system: the same phase space, but the rule changes in time – the same state can lead to different futures. With slowly changing parameters, the fixed points and cycles of the frozen system move (slide 10).

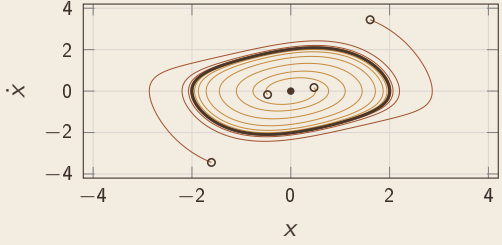
Fixed points and limit cycles

Problem: where do trajectories go in the long run?

Mass–spring–damper (L2): to a fixed point



van der Pol, $\mu = 0.3$: to a limit cycle



Fixed point: a state that does not change, $\dot{x} = 0$: for $(x, \dot{x})$ both $\dot{x} = 0$ and $\ddot{x} = 0$. **Limit cycle**: an isolated closed trajectory. **Attractor**: the set of states that trajectories approach in the long run – left a *point attractor*, right a *cycle attractor*. Open circles: starts. Right: from any start except the fixed point at the origin, the trajectory ends on the same cycle; its size and shape are set by the system. Larger μ: a distorted cycle ($\mu = 2$, slide 4), the same behaviour.

Data: RK4: mass–spring–damper ($c = 0.5$, $k = 4$, $m = 1$); van der Pol ($\mu = 0.3$).

Reminder: eigenvalues and eigenvectors

Problem: for a linear system $\dot{\mathrm{x}} = A\mathrm{x}$, how do we know whether it decays, grows or oscillates without solving it?

An **eigenvector** v of A is a direction that A only stretches: $A\mathrm{v} = \lambda\mathrm{v}$, with the **eigenvalue** λ (real or complex). Along it, $\dot{\mathrm{x}} = \lambda\mathrm{x}$, so

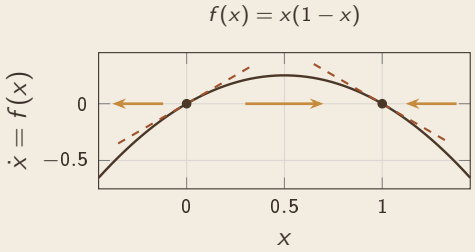
$$\boxed{\mathrm{x}(t) = \sum_i c_i\, e^{\lambda_i t}\, \mathrm{v}_i} \qquad (c_i \text{ from the initial condition}).$$

Eigenvalue $\lambda = \alpha + i\omega$	Motion along v
$\alpha < 0$	decays at rate $\lvert\alpha\rvert$
$\alpha > 0$	grows: the state leaves
$\omega \neq 0$	oscillates at angular frequency ω

A nonlinear system near a fixed point: the next slide.

Near a fixed point: the Jacobian

Problem: van der Pol is nonlinear: $\dot{x} = f(x)$ has no matrix A. How can the eigenvalues still tell us the motion near a fixed point?



Dashed: the tangents at the fixed points. Arrows: the direction in which x moves.

One variable. Near a fixed point, $f(x^*) = 0$, replace f by its tangent, as in Newton's method (L3): $f(x) \approx f'(x^*)\,(x - x^*)$. With $\delta = x - x^*$: $\dot{\delta} = f'(x^*)\,\delta$, so $\delta(t) = \delta_0\, e^{f'(x^*)\,t}$. At $x^* = 0$, $f' = 1$: the state leaves. At $x^* = 1$, $f' = -1$: it returns.

Several variables. Each f_i has a slope in each direction: the **Jacobian** $J_{ij} = \partial f_i / \partial x_j$, taken at x^*. Then $\dot{\boldsymbol{\delta}} = J\boldsymbol{\delta}$: the linear system of slide 7, with $A = J$.

Van der Pol: $f_1 = v$, $f_2 = \mu(1 - x^2)v - x$: $\quad J = \begin{pmatrix} 0 & 1 \\ -2\mu x v - 1 & \mu(1 - x^2) \end{pmatrix}$, $\quad J(0,0) = \begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix}$.

Valid only close to x^*: the terms left out grow as $|\boldsymbol{\delta}|^2$.

Eigenvalues at the two fixed points

Problem: do the eigenvalues predict the motion we saw on slide 6?

Mass–spring–damper, $c = 0.5$, $k = 4$

$\dot{x} = v$, $\dot{v} = -kx - cv$:

$$A = \begin{pmatrix} 0 & 1 \\ -4 & -0.5 \end{pmatrix}, \quad \lambda = -0.25 \pm 1.98\, i$$

Decays at rate $0.25\,\mathrm{s}^{-1}$, oscillates at $\omega = 1.98$ rad/s: exactly $\omega_d = \sqrt{k/m - (c/2m)^2}$ of L2. Negative real parts: the fixed point is an **attractor** (the inward spiral).

van der Pol at the origin

Jacobian at $(0, 0)$ (slide 8):

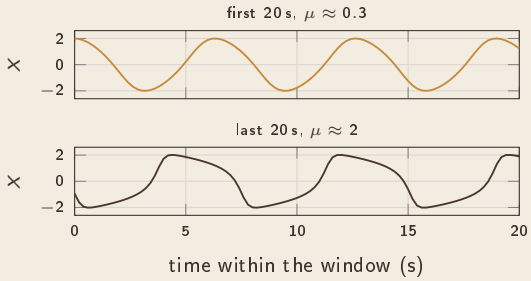
$$J = \begin{pmatrix} 0 & 1 \\ -1 & \mu \end{pmatrix}, \quad \lambda = \tfrac{1}{2}\left(\mu \pm \sqrt{\mu^2 - 4}\right)$$

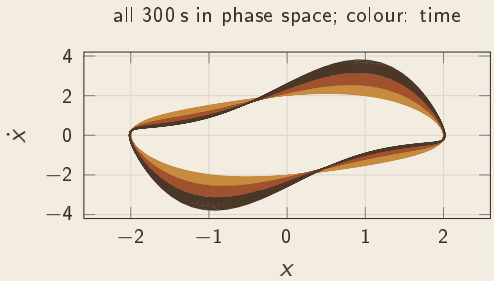
$\mu = 0.3$: $\lambda = 0.15 \pm 0.99\, i$, grows while oscillating (slide 6). $\mu = 2$: $\lambda = 1, 1$, leaves without oscillating. Positive real parts: the origin repels; the state settles on the limit cycle, as for $|x| > 1$ the damping turns positive.

Eigenvalues give the local motion near a fixed point; the limit cycle is a global, nonlinear property.

When the parameters change in time

Problem: what does phase space show for a nonautonomous system (L2)?





μ rises slowly from 0.3 to 2. Same phase space, but the rule changes: the cycle changes shape (largest $\dot{x}$ 2.5 → 3.8, period 6.4 → 7.6 s). Right (ochre: 0–100 s, red: 100–200 s, dark: 200–300 s): no single closed curve – states from different times are mixed.

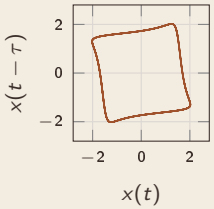
Phase space suits autonomous dynamics; nonautonomous change must be followed in time (L6).

Data: van der Pol with $\mu(t) = 0.3 + 1.7\,t/300\,\mathrm{s}$, RK4, 300 s.

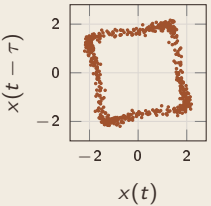
Phase space from one measured series: delay coordinates

Problem: often only $x(t)$ is measured, and $\dot{x}$ from data is noisy (L4). Can we still draw the phase space?

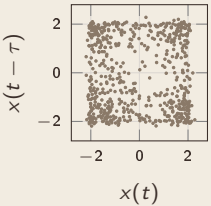
without noise



with noise, $\sigma_\varepsilon = 0.1$



the same values, shuffled



Delay coordinates: plot $x(t)$ against $x(t-\tau)$. A common starting point for τ: the first zero of the autocorrelation (L6), here 1.95 s $\approx P/4$; nearby τ give much the same shape. More complex systems: more delays, $(x(t), x(t-\tau), x(t-2\tau), \dots)$. Right: the noisy values in random order – same mean, s and histogram, no ring. Noise blurs the ring; shuffling destroys it: the shape is in the order of the values.

Data: van der Pol of slide 4, 200 s; centre: plus independent Gaussian noise ε, $\sigma_\varepsilon = 0.1$; right: those values shuffled.

PART B

Decomposing data

How variables vary together, and along which directions: correlation, the singular value decomposition and principal components – for independent measurements (rows: people, samples, repeated experiments) and for time series (rows: successive moments, or delay vectors).

The algebra is the same for both; for a time series, neighbouring rows are not independent, and their order carries information.

Covariance and correlation

Problem: the variance (L5) describes how one quantity varies. How do two quantities vary together?

Covariance: N pairs (x_i, y_i), each measured on the same unit or at the same moment:

$$s_{xy} = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})$$

The variance extended to two variables: $s_{xx} = s_x^2$. Positive when x and y tend to be on the same side of their means, negative when on opposite sides. Units of x times units of y, so its size depends on the units.

Correlation (Pearson): $r = s_{xy}/(s_x s_y)$, between -1 and 1, no units. It measures how close the points lie to a *straight line*.

Rank correlation (Spearman), r_S: r of the ranks (L5): a *monotonic* relation; robust to bad values.

Many variables: covariance matrix

Data matrix X: one row per observation (a unit, or a moment of a series), one column per variable. For m variables: $C_{jk} = s_{x_j x_k}$; on the diagonal, the variances. With centred columns, $C = X^{\mathsf{T}}X/(N-1)$.

Example, noisy van der Pol (slide 11), x_n and x_{n+4}, 0.2 s apart:

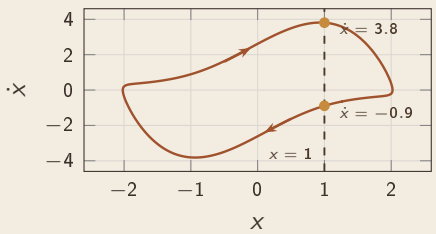
$$C = \begin{pmatrix} 2.217 & 2.165 \\ 2.165 & 2.217 \end{pmatrix}, \quad r = 0.976, \quad r_S = 0.965$$

Data: the noisy van der Pol series of slide 11.

Correlation versus dependence

Problem: does $r = 0$ mean unrelated?

Example: van der Pol limit cycle



Here $r = 0.00$ ($r_S = -0.14$), yet at a given x (dashed line) $\dot{x}$ takes only two values (dots): zero correlation, strong dependence.

- **What r tells reliably:** how strongly two quantities rise together, or one rises as the other falls (r_S: also along a curve, as long as it keeps going one way).
- **What it cannot tell:** whether they are related at all: $r = 0$ *can* hide a strong dependence, as in the example; what causes what (both may follow a third variable, e.g. time); **coupling**, the interaction in the dynamics (L8).
- **So:** plot the pair before reading r.

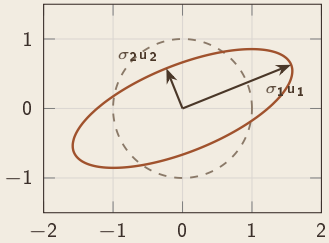
Any dependence: e.g. the **mutual information**, zero only for independent variables (for reference).

Data: van der Pol of slide 4, 200 s, without noise.

The singular value decomposition (SVD)

Problem: the points of slide 11 spread mostly along one straight direction ($r = 0.976$, slide 13). With many variables, how do we find these directions from the data table X itself? X is $N \times m$, not square, so the eigenvectors of slide 7 do not apply.

A maps the unit circle to an ellipse



Every $m \times n$ matrix: $\boxed{A = U\Sigma V^{\mathsf{T}}}$

- V: orthonormal (perpendicular, unit-length) input directions v_i
- Σ: **singular values** $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$, the stretches
- U: orthonormal output directions u_i

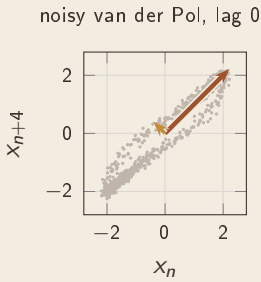
$Av_i = \sigma_i u_i$: a rotation, a stretch, a rotation.

What the σ_i tell:

- rank: the number of nonzero σ_i
- least squares: `lstsq` (L3) works through the SVD
- keep the k largest: the closest matrix of rank k (the step to PCA)

Principal component analysis (PCA)

Problem: many variables vary together. Along which directions do the data really vary?



Pairs (x_n, x_{n+4}), 4 samples = 0.2 s apart: so close in time that the two values nearly agree, and the cloud is long and thin. (At $\tau = 1.95$ s of slide 11 it is a ring, with no preferred direction.) Direction 1 (red, diagonal): 98.8% of the variance; 2 (ochre): 1.2%. Arrows: 1.5 standard deviations along 1, 2.5 along 2.

Data: the noisy van der Pol series of slide 11.

Data matrix X (slide 13), $N \times m$, centred. Its SVD: $X = U\Sigma V^{\mathsf{T}}$

- **principal directions**: the v_k
- variance along v_k: $\sigma_k^2/(N-1)$
- **principal components**: $XV = U\Sigma$, one value per row – for a series, a time course

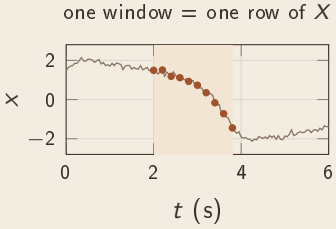
Linear statistics only: PCA uses only means, variances and covariances; equal covariances give the same PCA. Each component is a linear combination of the variables.

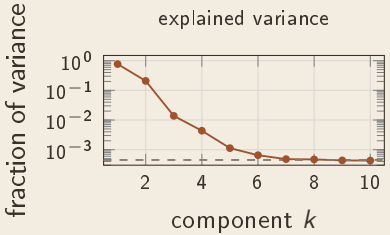
Row order ignored: shuffled rows give the same result; for a series, order enters only via delay coordinates.

Each component = a fixed pattern over the variables (v_k) × its profile over the rows (u_k).

How many components? Ten delay coordinates

Problem: slide 16 used two delay coordinates. With more (slide 11), each row becomes a **window** of the record (left): 10 values 0.2 s apart, spanning 1.8 s; the window slides by one sample from row to row. PCA then gives 10 directions. How many hold the signal, and how many only noise?







Components 1–2: 77% and 21%; 3–6: from 1.4% down to 0.07%; 7–10: a flat floor (dashed): independent noise adds the same variance, $\sigma_\varepsilon^2 = 0.01$, to every direction. Above the floor: signal. With correlated noise the floor is not flat. Components 1–2 trace the limit cycle; 3–6 its non-sinusoidal shape (harmonics, L6). PCA of delay coordinates is called **singular spectrum analysis**.

Data: the noisy van der Pol series of slide 11, 4001 values.

What PCA does and does not tell

Problem: PCA always returns directions. What can we read from them, and what not?

	PCA gives	**but not**
Shape	straight directions, from the covariances alone	curved structure: a closed curve needs two components; equal covariances give the same PCA
Units	directions in the units you chose	a unit-free answer: for mixed units, **standardise** (subtract each mean, divide by its s)
Signal	how many directions are above the noise floor: a lower-dimensional picture	what they are: components are only uncorrelated, not independent causes
Ranking	the order of variance	the order of importance: a weak rhythm can have small variance
Rows	the same result for any table of rows	their order: in a series, only delay coordinates carry it

Before reading any PCA: check the units and the noise floor (slide 17).

PART C

Outlook

Other decompositions, and the systems they suit.

Decompositions and the systems they suit

Problem: which decomposition suits which data?

Method	Idea	Assumes
PCA (today)	fixed patterns over the variables, ordered by variance	covariances only (straight structure); one set of directions for the whole record
Singular spectrum analysis (today)	PCA of sliding windows (delay coordinates) of one series	a stationary series
Dynamic mode decomposition	fit $x_{n+1} = Ax_n$ to many measured variables; each eigenvector of A is a mode with its own frequency and growth or decay rate (slide 7)	linear, autonomous dynamics
Independent component analysis	unmix signals that are fixed mixtures of independent sources	independent, non-Gaussian sources; fixed mixing
Nonlinear mode decomposition	extract each oscillation from a time–frequency view (as the STFT, L6), keeping its harmonics in one mode; surrogates decide oscillation or noise	oscillations; frequency and amplitude may change in time
Clustering (L10)	groups of points: k-means, DBSCAN	round groups, or dense regions

Which assumption fails first?

Problem: the system types of Lectures 1 and 2 – which decompositions does each break?

System (L1, L2)	What goes wrong	Better suited
linear, autonomous, stationary (e.g. noise-driven)	little: the same straight directions throughout	PCA, SSA, DMD
linear, autonomous, decaying or growing	PCA mixes early and late	DMD (growth, decay rates)
nonlinear	a curve needs several straight directions; harmonics split into separate modes	phase space; NMD
nonautonomous	one fixed set of directions or frequencies averages the change	NMD; time-resolved views (L6)
stochastic, noisy	independent noise adds a floor to every component (slide 17)	components above the floor; surrogates
switching states	PCA mixes the states into one cloud	clustering (L10)

Classify the system first (L2): the classification tells which decomposition to trust.

Which method for which question?

Question	Method	Assumes	Check
What states does it visit?	phase portrait; delay coordinates	enough coordinates; good τ	shuffled surrogate; vary τ
Do two variables vary together?	r; rank r_S	one tendency: straight (r) or monotonic (r_S)	plot the pair: $r = 0 \not\Rightarrow$ unrelated; p-value only for independent pairs
Local motion at a fixed point	eigenvalues of the Jacobian J (slide 8)	close to the point	simulate
Main directions of variation	PCA (SVD of centred data)	covariances only; one set of directions; comparable units	noise floor (slide 17); for mixed units, standardise (slide 18)

Before you decompose: four questions

1. What is the state of the system, and do I measure it, or must I reconstruct it? *(delay coordinates)*
2. Is the structure in the order of the values? *(compare with a shuffled surrogate)*
3. How many components stand above the noise floor, is the structure straight or curved, and are the units comparable? *(PCA)*
4. Does the correlation fit the relation? *($r = 0$ does not mean unrelated: plot the pair)*

Plot the points first; then choose the decomposition.

TO TAKE AWAY

In phase space a rhythm is a closed curve.

Correlation and PCA are linear statistics: built from means, variances and covariances, they describe straight-line structure. They tell reliably whether quantities rise and fall together, and along which straight directions the data spread.

They can be applied as they stand when the variation lies along straight directions that stay the same through the data (for a time series: a stationary system), with variables in the same units or standardised.

Otherwise they still return numbers, read differently: a curve (a limit cycle) needs several straight components, and r can be zero despite a strong dependence; a change in time gives one average set of directions; switching states give one mixed cloud. So plot the data first.